Pertussis and the CMI-PB project (Class 18)

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Background

Pertussis (whooping cough) is a common lung infection caused by the bacteria B. pertussis. It can infect anyone but it is most deadly for infants (under 1 year of age).

CDC tracking data

The CDC tracks the number of Pertussis cases:

```
cdc
```

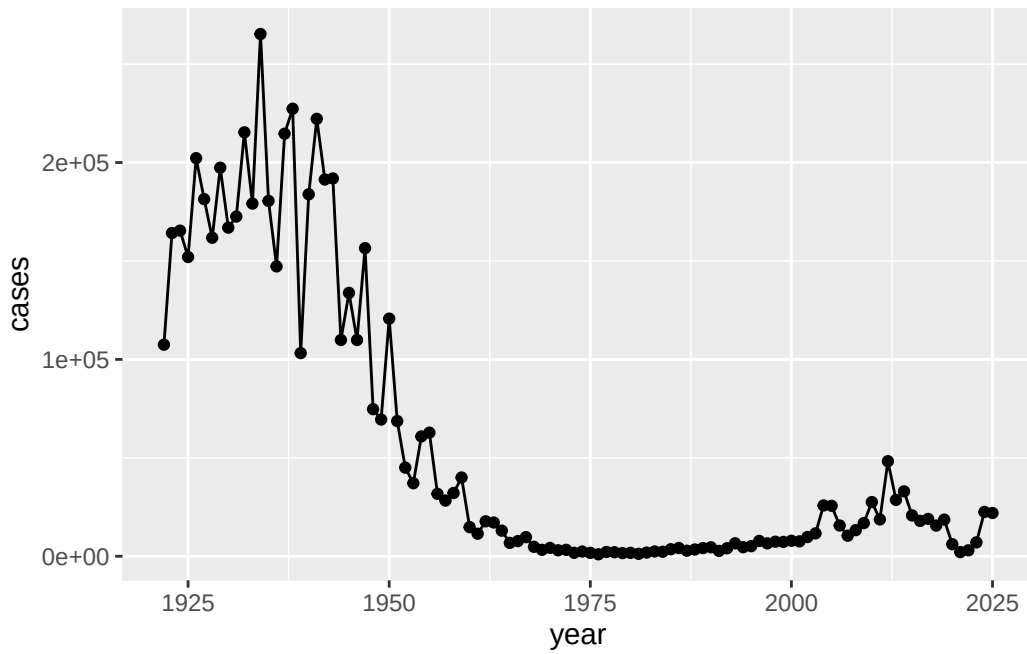
```
    year  cases
1   1922 107473
2   1923 164191
3   1924 165418
4   1925 152003
5   1926 202210
6   1927 181411
7   1928 161799
8   1929 197371
9   1930 166914
10  1931 172559
11  1932 215343
12  1933 179135
13  1934 265269
14  1935 180518
15  1936 147237
16  1937 214652
17  1938 227319
18  1939 103188
```

19	1940	183866
20	1941	222202
21	1942	191383
22	1943	191890
23	1944	109873
24	1945	133792
25	1946	109860
26	1947	156517
27	1948	74715
28	1949	69479
29	1950	120718
30	1951	68687
31	1952	45030
32	1953	37129
33	1954	60886
34	1955	62786
35	1956	31732
36	1957	28295
37	1958	32148
38	1959	40005
39	1960	14809
40	1961	11468
41	1962	17749
42	1963	17135
43	1964	13005
44	1965	6799
45	1966	7717
46	1967	9718
47	1968	4810
48	1969	3285
49	1970	4249
50	1971	3036
51	1972	3287
52	1973	1759
53	1974	2402
54	1975	1738
55	1976	1010
56	1977	2177
57	1978	2063
58	1979	1623
59	1980	1730
60	1981	1248
61	1982	1895

62	1983	2463
63	1984	2276
64	1985	3589
65	1986	4195
66	1987	2823
67	1988	3450
68	1989	4157
69	1990	4570
70	1991	2719
71	1992	4083
72	1993	6586
73	1994	4617
74	1995	5137
75	1996	7796
76	1997	6564
77	1998	7405
78	1999	7298
79	2000	7867
80	2001	7580
81	2002	9771
82	2003	11647
83	2004	25827
84	2005	25616
85	2006	15632
86	2007	10454
87	2008	13278
88	2009	16858
89	2010	27550
90	2011	18719
91	2012	48277
92	2013	28639
93	2014	32971
94	2015	20762
95	2016	17972
96	2017	18975
97	2018	15609
98	2019	18617
99	2020	6124
100	2021	2116
101	2022	3044
102	2023	7063
103	2024	22538
104	2025	21996

Q.I want a plot of year on the x vs cases on the y.

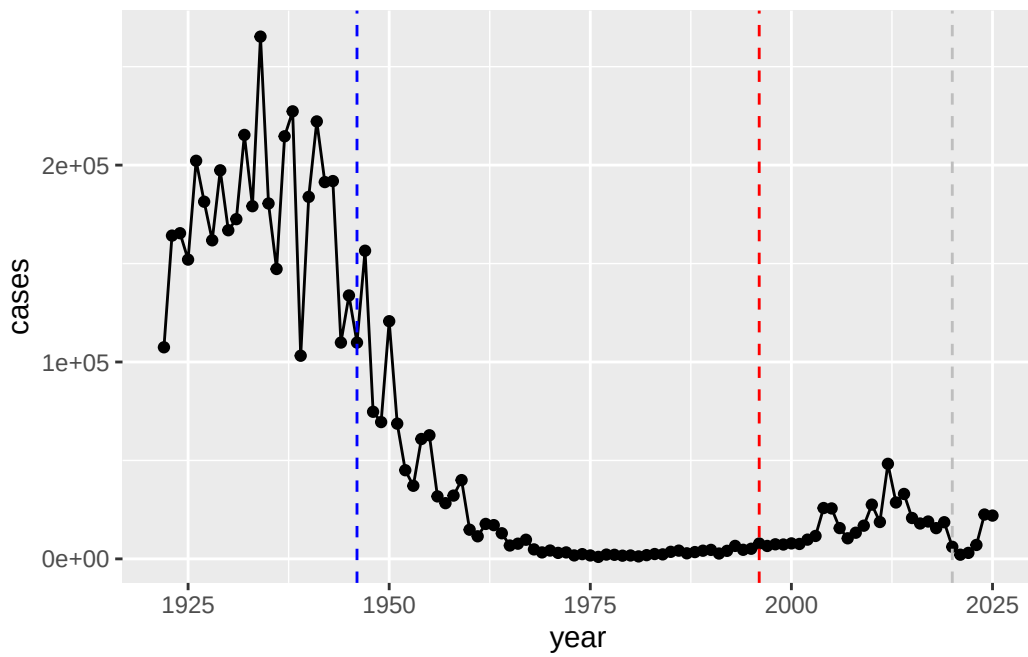
```
library(ggplot2)

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Q. Add annotation lines for the major milestones of wP vaccination roll-out (1946) and the switch to the aP vaccine (1996). What do you notice?

We see that there is an increase past 2020.

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty =2) +
  geom_vline(xintercept = 1996, col="red", lty=2) +
  geom_vline(xintercept = 2020, col="gray", lty=2)
```



> Q2. Using the ggplot geom_vline() function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

I notice a sharp decline in cases from the introduction of the wP vaccine to the introduction of the 1996 aP vaccine. In 1996 when the new aP vaccine was introduced, we see cases nearly at zero.

> Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine, there is a steady increase after 2000 with a sharp peak around ~2010. However, there is again a steady increase after 2020 due to vaccine hesitancy, antibiotic overuse and AMR. This can also suggest that there is a waning efficacy of the aP vaccine is lower than that of the wP vaccine, thus requiring booster vaccinations for the aP.

Exploring CMI-PB data

The CMI-PB project link: < https://www.cmi-pb.org >

The project's mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Basically, make available a large dataset on the immune response to Pertussis. They use a "booster" vaccination as a proxy for Pertussis infection.

They make their data available as JSON format API. We can read this into R with the `read_json()` function from the **jasonlite** package:

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject",
                     simplifyVector = TRUE)
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many aP and wP individuals are there in this `subject` table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. How many male/female?

```
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. What is the bread down of biological sex and race? Q. Is this representative of the US population?

```
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

We can now read more tables from the CMI-PB database

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer",
                      simplifyVector = TRUE)
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To make sense of all this data, we need to "join" (a.k.a "merge" or "link") all these tables together. Only then will you know that a given antibody measurement (from the `ab_titer` table) was collected on a certain data from a certain wP or aP subject (from the `subject` table).

We can use **dplyr** and the `*_join()` family of functions to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Let's do one more `inner_join()` to join the `ab_titer` with all our `meta` data.

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
```

```
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. How many different Ab "isotypes" values are in this dataset

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q 12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

We see that there is there is double the amount of values in the 2023 dataset then there are in the 2022 dataset.

Q. How many different "antigen" values are measured

```
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
    PD1     PRN      PT     PTM   Total      TT
   1970    6712    6712    1970     788    6318
```

Let's focus on IgG isotype

Note: there are two ways to write the pipe command; |> or %>%

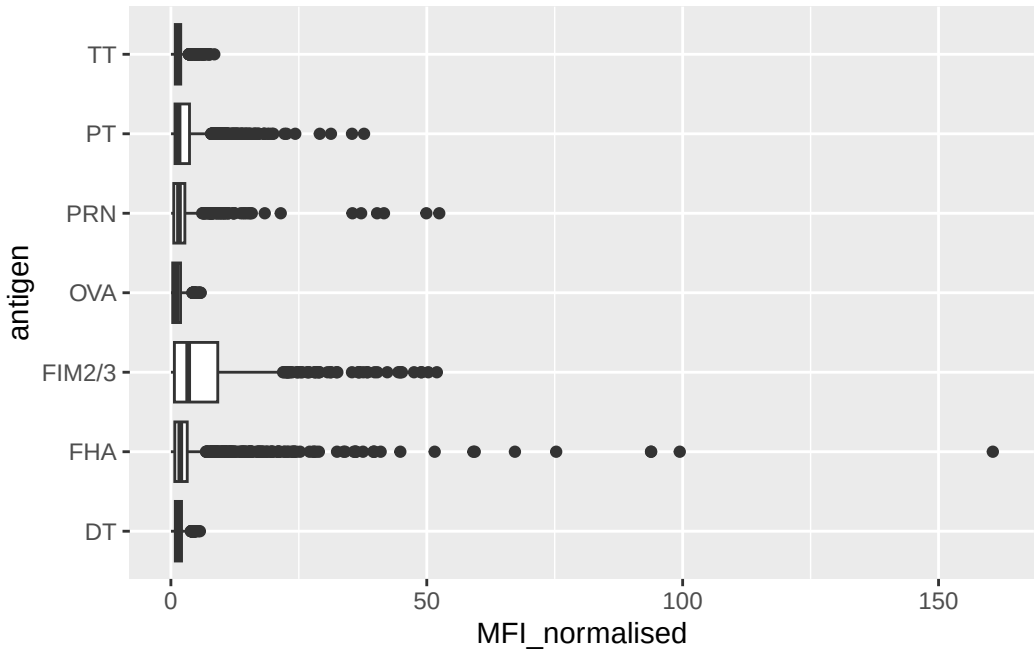
```
igg <-abdata |>
      filter(isotype=="IgG")

head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
```

```
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Make a plot of `MFI_normalised` values for all `antigen` values.

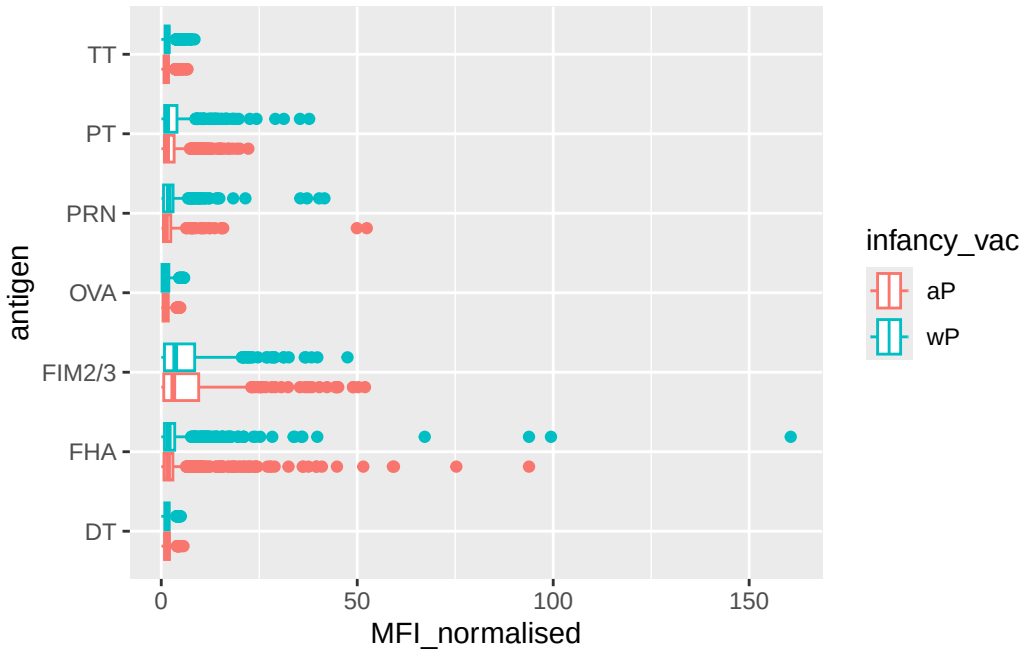
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



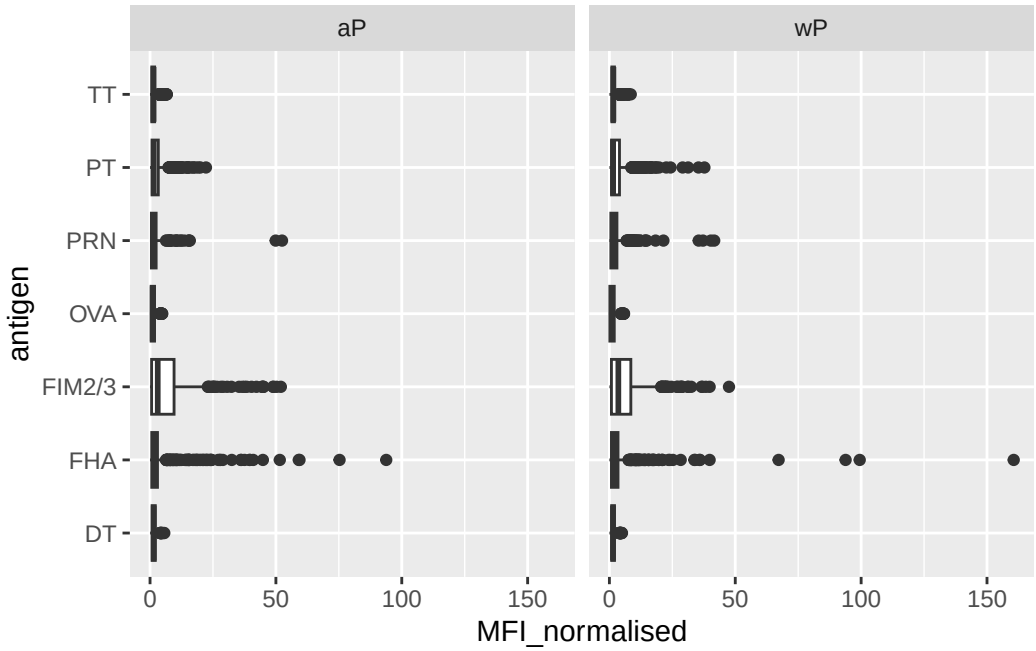
The antigens "PT", "FIM2/3" and "FHA" appear to have the widest range because the aP vaccines is composed of these antibodies.

Q. IS there a difference for these responses between aP and wP individuals.

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



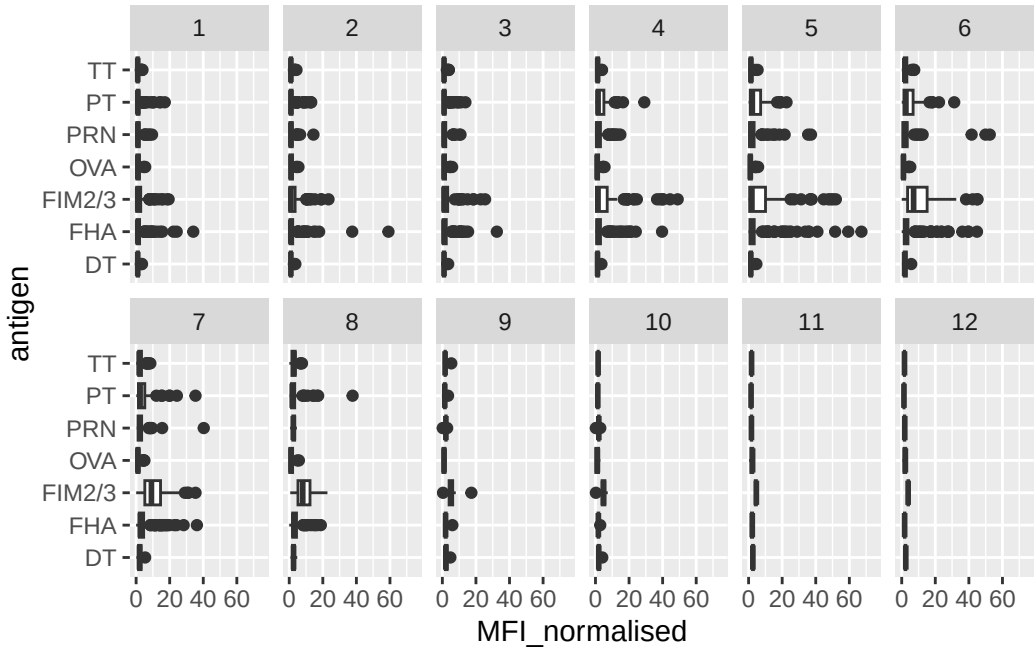

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Q. 13 Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

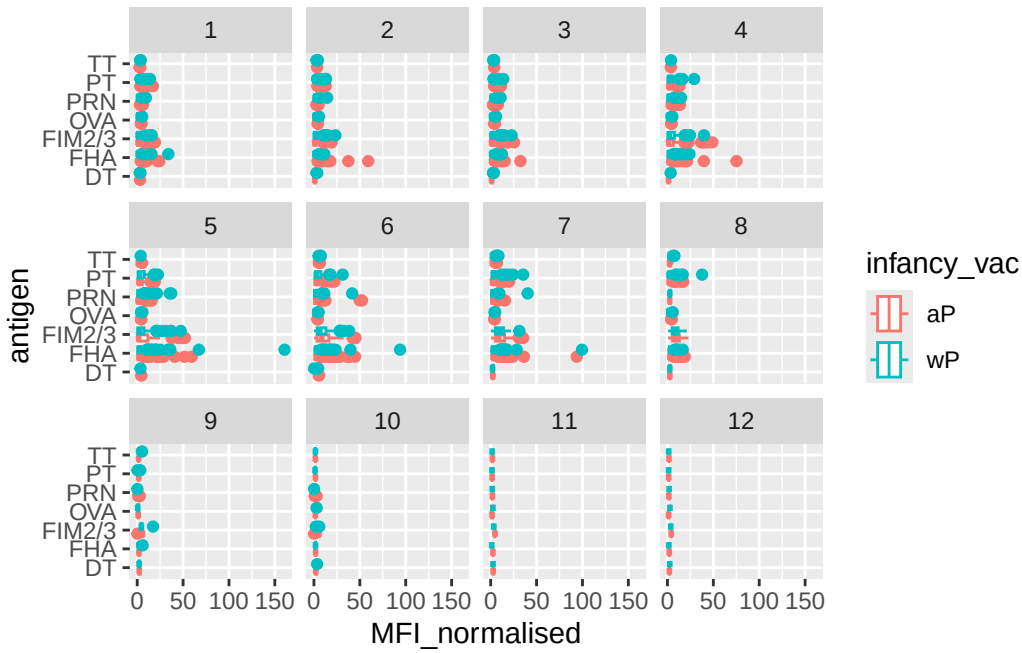


Q. 14 What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens that show the most obvious differences in IgG antibody levels over time are FIM2/3 and FHA. FIM2/3, which are the antigen components the whooping cough bacterium, recognition increases over time due to a boosted antigen-specific IgG response after vaccination.

Q. Is there a difference with time (i.e. before booster shot vs after booster shot)

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```

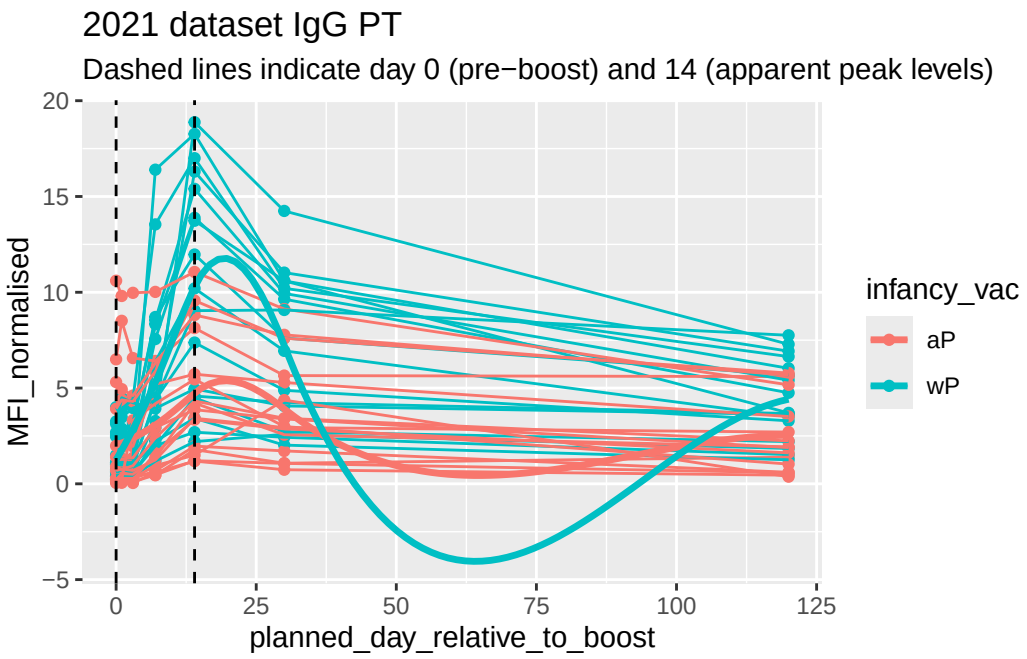


Let's look specifically at 2021 IgA in wP and aP

```
## filter to 2021 dataset, IgG and PT only
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG",  antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
     geom_smooth(
    aes(group = infancy_vac, col = infancy_vac),
    method = "loess",  # or method = "lm"
    se = FALSE,
    linewidth = 1.2
  )+
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
```

```
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



ADD A TREND LINE FOR WP and AP seperately